Summary of OG-Core macroeconomic model
For linking OG-Core to the CLEWS model

This document is organized into the following sections

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1. Links to the model code and documentation online

All the code for OG-Core is available at https://github.com/PSLmodels/OG-Core.

The online documentation for the model, including description of the Python API as well as the theory behind the code, is available at https://pslmodels.github.io/OG-Core.

The model can also be directly downloaded from the Python Package Index (PyPI.org) as a Python package `ogcore`, which can be directly installed on your local machine using the `pip install ogcore` command.

2. Summary of how the model works

OG-Core is a large-scale open source dynamic general equilibrium model of country fiscal policy. This model solves for the equilibrium outcomes of an economy from the current period until a date in the distant future. The model first solves for the very long run economic outcomes, or steady-state. The model solves for the equilibrium transition path of the economic outcomes from the current period to a period in the distant future when the economy reaches its steady state.

In the most common calibration of OG-Core, equilibrium in each period is characterized by 1,120 nonlinear equations and unknown variables (2 household decisions x 80 ages x 7 ability types). Because we solve the model for up to 320 time periods (years), the equilibrium outcomes of the model are characterized by a system of more than 358,400 nonlinear equations.

These equations represent the optimal decisions of businesses and households over their lifetimes, subject to budget constraints and government activity. The model has very detailed demographic dynamics, modeling how the population distribution by age evolves over time using mortality rates, fertility rates, and immigration rates by age. OG-Core also accurately reflects the interactions of government with households (through taxes and transfers) and with businesses (through taxes and infrastructure investment). Finally, the model includes financial flows of capital from abroad.

3. Summary of OG-Core model inputs that could come from CLEWS outputs

A full list of the inputs to OG-Core is available in the online documentation in an appendix chapter entitled, "Model parameters". These are the primary factors that the CLEWS model could inform and feed into.

- Carbon intensity by production sector (useful for analyzing a carbon tax)
- Mix of factors to produce goods: e.g., mix of coal and gas in energy production, that can be used to update the input-output (IO) matrix in the OG model
  - Depends on level of detail in IO matrix
  - CLEWs can describe how gov't policy may change the mix of production like this -- and again this can be used to affect the IO matrix in OG
- Not directly in OG, but could help with analysis from OG:
  - Land, water, and other resource use
- Tom says important to align on the investment flows
  - e.g., carbon tax imposed, in CLEWs this would spur investment in new green tech, but in OG, this increases cost of energy and would reduce investment
    - How would these be aligned?
    - Can you do something to represent this increase in investment? Are the economics really working out in the way Tom suggests-- would investment actually increase when costs are higher? Doesn't seem right, but maybe there's something in the dynamics
    - Maybe the thing is that we need different types of capital (e.g., green and dirty capital)
      - Or is differentiated industries enough (e.g, green energy and dirty energy and the relative prices change)?

4. Summary of OG-Core model output that could inform CLEWS inputs

These outputs are the primary objects that could inform and feed into the CLEWS model. Formally, the output of the OG-Core model is summarized in the respective return objects (Python dictionaries, both named "`output`") of the `run_SS()` function of the `SS.py` module and of the `run_TPI()` function of the `TPI.py` module.

- Investment, output by production sector (specificity of sector depends on data)

- Consumption by type of good (e.g., food, durables, services, etc -- can get more specific if needed)
    - These demands can be mapped to CLEWs demands
- Population
- Maybe prices for commodities used in resource production
    - e.g., cost of steel for coal power is determined by demands in the goods market
    - And interest rates matter

5. Types of research questions

OG-Core country calibrations are good at answering the following types of questions.

- What is the effect of:
    - fiscal policy changes
    - demographic changes
- On
    - Macroeconomic variables
        - GDP, Debt, debt-to-GDP, employment, aggregate capital, aggregate investment, aggregate consumption, interest rates, wages, government revenue, deficits
    - Household/distributional economic outcomes
        - Consumption, before-tax income, after-tax income, net tax liability, labor supply, savings, wealth, government transfers, government pensions.

6. References to papers using OG-Core

We have an appendix chapter in the online documentation for OG-Core, "Citations and use cases of OG-Core," that lists all the references to studies that we know of that have used the model for policy or research. The list includes 17 references. We highlight two of them here.

- Jason DeBacker, Richard W. Evans, and Benjamin R. Page. "**A detailed macroeconomic analysis of President Biden's 2020 campaign tax proposals**," Working Paper, Tax Policy Center, July 2021. This paper uses the OG-USA calibration of OG-Core to estimate the impact of some of President Biden's campaign proposals from 2020 on the distribution of households' economic outcomes and on the macroeconomy.

- Jason DeBacker, Richard W. Evans, and Raiany Romanni, "**The Macroeconomics of Improving Aging**," forthcoming. This paper uses the OG-USA calibration of OG-Core. We use the demographics inputs to the model as well as the productivity profiles to quantify the effects of medical breakthroughs of reducing the effects of aging and their effect on economic output over the long run.